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COURSE: CHE 312: Transport Phenomena II

Spring Semester 2020

Department of Chemical Engineering

University of Illinois at Chicago

Midsemester Exam #2

Only one sheet of paper 5 x 7 is allowed

Duration: 55 minutes --- March 9, 2020

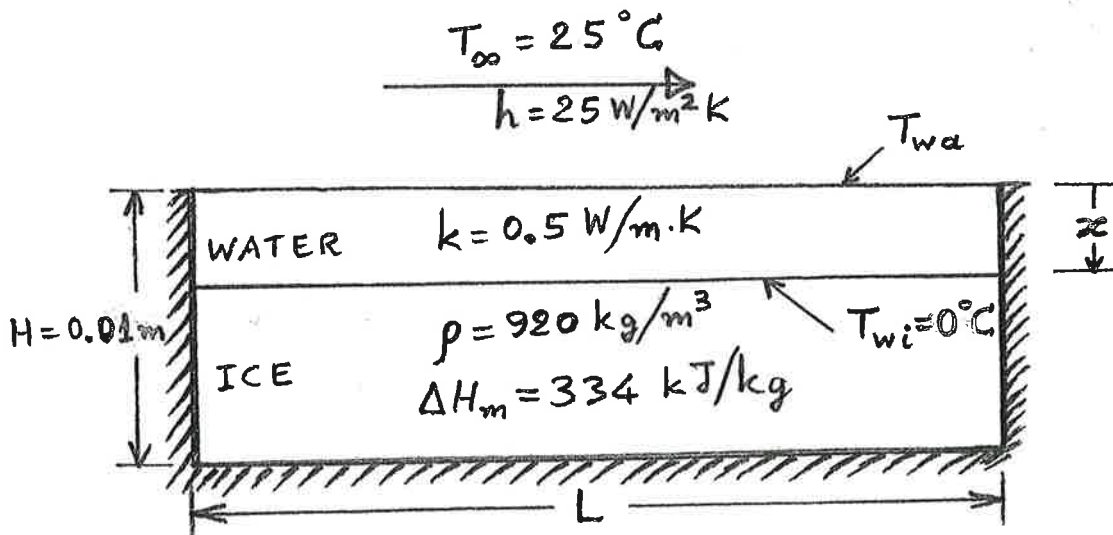
40/125
+5
=45

Problem 1: (50 points)

A slab of ice in a container of dimensions L , L , and H is placed on a well-insulated pad. At its top surface, initially the ice is exposed to ambient air for which $T_\infty = 25^\circ\text{C}$ and the convection coefficient is $25 \text{ W/m}^2\cdot\text{K}$. The ice is melting and the water formed acts as an additional resistance to the heat transfer. Assume that there is no storage of energy in the water being formed and that the water temperature profile is linear. The ice temperature remains at 0°C and $H = 10 \text{ mm}$. If the density and latent heat of fusion of ice are 920 kg/m^3 and 334 kJ/kg , respectively, and the thermal conductivity of water is $0.5 \text{ W/m}\cdot\text{K}$, then:

- By performing energy balance(s) over appropriate control element(s) determine the position of the water/ice interface as a function of time.
- Determine the time required for the ice to melt. How does this relate to the time found in the case of no water-layer resistance?

Bonus: (25 points) Determine how the water-air interface temperature, T_{wa} , changes with time. What is the value of T_{wa} when the ice has melted?



(10/50)

known $T_{\infty} = 25^{\circ}\text{C}$ $h = 25 \text{ W/m}^2\text{K}$ $H = 0.01 \text{ m}$ $T = 0^{\circ}\text{C}$ $\rho = 920 \text{ kg/m}^3$
 $A_{\text{film}} = 334 \text{ kJ/kg}$ $k = 0.5 \text{ W/mK}$

a) $-hA(T - T_{\infty}) = \rho V c \frac{dT}{dt}$ $T_{\text{film}} = \text{linear}$

$$x(t) = \sqrt{\frac{\alpha t}{Fo}}$$

$$-hA(T - T_{\infty}) = \rho V c \frac{dT}{dt}$$

$$-hA \Delta T = \rho V c \ln\left(\frac{T_{\text{film}} - T_{\infty}}{T_{\text{film}} - T_0}\right)$$

$$t = -\frac{\rho V c}{hA} \ln\left(\frac{T_{\text{film}} - T_{\infty}}{T_{\text{film}} - T_0}\right)$$

$$x(t) = \sqrt{\frac{\alpha t}{Fo}}$$

dimensionless units

$$-k \frac{dT}{dh} \Big|_{h=L} = -h(T - T_{\infty})$$

$$-hA \Delta T = \rho V c \ln\left(\frac{T - T_{\infty}}{T_0 - T_{\infty}}\right)$$

$$h \frac{-A \Delta T}{\rho V c} = \ln\left(\frac{T - T_{\infty}}{T_0 - T_{\infty}}\right)$$

$$\frac{-A \Delta T}{\rho V c} (T - T_{\infty}) = \frac{dT}{dh}$$

$$\frac{A \Delta T}{\rho V c} (T - T_{\infty}) k = 1$$

$$t = \frac{\rho V c}{A(T - T_{\infty})k}$$

$$Fo = \frac{\alpha t}{L^2}$$

$$Bi = \frac{hL}{k}$$

$$\alpha = \frac{k}{\rho c}$$

$$\theta^* = \frac{\theta}{\theta_i} = \frac{T - T_{\infty}}{T_0 - T_{\infty}} \quad x^* = \frac{x}{L}$$

$$t^* = \frac{\alpha t}{L^2} = Fo$$

$$Bi = \frac{(25 \frac{\text{W}}{\text{m}^2\text{K}})(0.01 \text{ m})}{0.5 \frac{\text{W}}{\text{mK}}} =$$

$$= 0.5$$

$$Bi > 0.1$$

~~Lumped model~~

b) for ice to completely melt $x = 0.01 \text{ m}$

$$x = 0.01$$

$$0.01 = \sqrt{\frac{\alpha t}{Fo}}$$

$$0.01$$

X

since water does not store energy, we can ignore water and act if ice and air are only system

X

Bonus: T_{film} stays constant when ice melts, when ice is completely done melting T_{film} will slowly increase as time increase

X

Problem 2: (75 points)

A fluid at 100 °C is to be cooled by a different stream of the same fluid at 20 °C. A heat exchanger is available for parallel or counter flow. When the heat exchanger is used with parallel flow, the outlet temperature of the hot stream is 65 °C. If the mass flow rates in the hot and cold streams are equal to each other and the heat capacity of the fluid is temperature independent, determine:

- (a) (30 points) The outlet temperature of the cold stream with parallel flow in the heat exchanger, and
 (b) (45 points) The outlet temperatures of the cold and hot streams when a counter flow set up is used.

parallel or counter flow

$$T_{hi} = 100^\circ\text{C} \quad T_{ci} = 20^\circ\text{C} \quad \dot{m}_c = \dot{m}_h \quad \text{Temp is independent}$$

$$T_{ho} = 65^\circ\text{C} \quad C_{ph} = C_{pc}$$

a) parallel outlet stream. outlet temp of cold stream in heat exchanger

$$dq = \dot{m}_h C_{ph} (dT_h) = - \dot{m}_c C_{pc} (dT_c)$$

$$Q = \dot{m}_h C_{ph} (T_{hi} - T_{ho}) = \dot{m}_c C_{pc} (T_{co} - T_{ci})$$

$$dq = \dot{m}_c C_{pc} dT_c = C_c dT_c$$

$$Q = UA \Delta T_{lm}$$

$$\int dq = \int \dot{m}_c C_{pc} (dT_c)$$

$$100^\circ\text{C} \quad 65^\circ\text{C}$$

$$Q = \dot{m}_h C_{ph} (T_{hi} - T_{ho}) \quad Q = \dot{m}_c C_{pc} (T_{co} - T_{ci})$$

$$20^\circ\text{C}$$

$$X = 55^\circ\text{C}$$

$$Q = \dot{m}_h C_{ph} (T_{hi} - T_{ho})$$

$$T_{co} - T_{ci} = T_{hi} - T_{ho}$$

$$X - 20 = 100 - 65$$

$$X = 55^\circ\text{C}$$

$$\epsilon = \frac{1 - \exp[-NTU(1+Cr)]}{1+Cr}$$

b) counter flow set up

$$100^\circ\text{C} \quad 65^\circ\text{C}$$

$$Q = \dot{m}_h C_{ph} (T_{hi} - T_{ho})$$

$$Q = \dot{m}_c C_{pc} (T_{co} - T_{ci})$$

$$X = 35^\circ\text{C}$$

$$X \quad 20^\circ\text{C}$$

NOT NECESSARILY NEEDED

TABLE 11.3 Heat exchanger effectiveness relations [5]

Flow Arrangement	Relation	
Parallel flow	$\varepsilon = \frac{1 - \exp[-NTU(1 + C_r)]}{1 + C_r}$	(11.28a)
Counterflow	$\varepsilon = \frac{1 - \exp[-NTU(1 - C_r)]}{1 - C_r \exp[-NTU(1 - C_r)]}$	$(C_r < 1)$
	$\varepsilon = \frac{NTU}{1 + NTU}$	$(C_r = 1)$ (11.29a)
Shell-and-tube		
One shell pass (2, 4, ... tube passes)	$\varepsilon_1 = 2 \left\{ 1 + C_r + (1 + C_r^2)^{1/2} \times \frac{1 + \exp[-(NTU)_1(1 + C_r^2)^{1/2}]}{1 - \exp[-(NTU)_1(1 + C_r^2)^{1/2}]} \right\}^{-1}$	(11.30a)
n shell passes ($2n, 4n, \dots$ tube passes)	$\varepsilon = \left[\left(\frac{1 - \varepsilon_1 C_r}{1 - \varepsilon_1} \right)^n - 1 \right] \left[\left(\frac{1 - \varepsilon_1 C_r}{1 - \varepsilon_1} \right)^n - C_r \right]^{-1}$	(11.31a)
Cross-flow (single pass)		
Both fluids unmixed	$\varepsilon = 1 - \exp \left[\left(\frac{1}{C_r} \right) (NTU)^{0.22} \{ \exp[-C_r (NTU)^{0.78}] - 1 \} \right]$	(11.32)
$C_{\max}$ (mixed), $C_{\min}$ (unmixed)	$\varepsilon = \left(\frac{1}{C_r} \right) (1 - \exp \{ -C_r [1 - \exp(-NTU)] \})$	(11.33a)
$C_{\min}$ (mixed), $C_{\max}$ (unmixed)	$\varepsilon = 1 - \exp(-C_r^{-1} \{ 1 - \exp[-C_r (NTU)] \})$	(11.34a)
All exchangers ($C_r = 0$)	$\varepsilon = 1 - \exp(-NTU)$	(11.35a)

TABLE 11.4 Heat exchanger NTU relations

Flow Arrangement	Relation	
Parallel flow	$NTU = -\frac{\ln[1 - \varepsilon(1 + C_r)]}{1 + C_r}$	(11.28b)
Counterflow	$NTU = \frac{1}{C_r - 1} \ln \left(\frac{\varepsilon - 1}{\varepsilon C_r - 1} \right)$	$(C_r < 1)$
	$NTU = \frac{\varepsilon}{1 - \varepsilon}$	$(C_r = 1)$ (11.29b)
Shell-and-tube		
One shell pass (2, 4, ... tube passes)	$(NTU)_1 = -(1 + C_r^2)^{-1/2} \ln \left(\frac{E - 1}{E + 1} \right)$	(11.30b)
	$E = \frac{2/\varepsilon_1 - (1 + C_r)}{(1 + C_r^2)^{1/2}}$	(11.30c)
n shell passes ($2n, 4n, \dots$ tube passes)	Use Equations 11.30b and 11.30c with $\varepsilon_1 = \frac{F - 1}{F - C_r} \quad F = \left(\frac{\varepsilon C_r - 1}{\varepsilon - 1} \right)^{1/n} \quad NTU = n(NTU)_1$	(11.31b, c, d)
Cross-flow (single pass)		
$C_{\max}$ (mixed), $C_{\min}$ (unmixed)	$NTU = -\ln \left[1 + \left(\frac{1}{C_r} \right) \ln(1 - \varepsilon C_r) \right]$	(11.33b)
$C_{\min}$ (mixed), $C_{\max}$ (unmixed)	$NTU = -\left(\frac{1}{C_r} \right) \ln [C_r \ln(1 - \varepsilon) + 1]$	(11.34b)
All exchangers ($C_r = 0$)	$NTU = -\ln(1 - \varepsilon)$	(11.35b)

